RADRAFT

AI Radiology Reporting Assistant

PROJECT DOCUMENTATION

TRACK DEPI — AI in Healthcare

COVERAGE Project Planning · Stakeholder Analysis · Database Design · UI/UX Design

EDITION Conversation-grounded MVP documentation

REVISED 31 October 2026

Clinical safety note: This educational MVP supports radiologists with draft preparation. Final clinical approval remains with the radiologist.

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Project: RADRAFT — AI Radiology Reporting Assistant

Track: DEPI — AI in Healthcare

Document purpose: Submission documentation covering Project Planning, Stakeholder Analysis, Database Design, and UI/UX Design.

Status: MVP / educational proof of concept. Clinical outputs require radiologist review and approval.

Evidence policy: Reconstructed from the project team's earlier design/build/debug discussions plus the uploaded interactive presentation. Distinguishes confirmed implementation, demonstrated workflow design, and proposed future work.

Revision: 31 October 2026 — conversation-grounded edition.

1. Project Overview

RADRAFT is an AI-assisted radiology reporting workflow designed to reduce the time radiologists spend converting imaging findings into formal reports. In the current MVP, the radiologist sends findings as text or voice through Telegram. The system authorizes the user, transcribes voice input when needed, applies privacy and safety checks, de-identifies protected health information, generates a structured draft using an AI agent with deterministic radiology calculators and a grounded RAG knowledge base, returns the draft to the radiologist for approval or adjustment, and finally renders and delivers a standalone HTML report.

The design follows a **human-in-the-loop** principle: the AI produces a draft, while the radiologist remains the final clinical authority and must approve the report before final delivery.

1.1 Current MVP Architecture

1. **Workflow 1 — Access & Input Gateway**

Telegram intake, bot/human check, PostgreSQL authorization, subscription-credit check, text/voice normalization, voice transcription, doctor-name retrieval.

2. **Workflow 2 — Privacy & Guardrails**

Harm/injection/topic checks, PHI detection, regex + Microsoft Presidio processing, de-identification and placeholder mapping.

3. **Workflow 3 — Clinical AI Agent & RAG**

LLM orchestration, deterministic calculator tools, Qdrant vector retrieval, Cohere reranking, grounded drafting and controlled re-identification.

4. **Workflow 4 — Human Review & Revision**

Draft delivery to the radiologist, approve/adjust loop, PHI protection on every edit, controlled report editing and versioning.

5. **Workflow 5 — Report Rendering & Delivery**

HTML formatting, trusted header/signature insertion, final Telegram delivery and usage-credit deduction trigger.

Support workflows: credit/subscription counting, administrative user provisioning, and medical knowledge-base ingestion.

1.2 Technology Stack

- **Workflow orchestration:** n8n running in Docker
- **User interface / MVP channel:** Telegram Bot API
- **Relational database:** PostgreSQL
- **Vector database:** Qdrant
- **PHI detection / de-identification:** Regex + Microsoft Presidio

- **LLM routing:** OpenRouter / Claude-family models in the current workflow design
- **Embedding model:** OpenAI embeddings
- **Reranking:** Cohere reranker
- **Voice transcription:** Gemini transcription
- **Local topic guard:** Ollama
- **Knowledge ingestion:** Google Drive → chunking → embeddings → Qdrant
- **Final report format:** standalone HTML

1.3 Product Scope Boundary

Implemented / demonstrated in the MVP: Telegram text and voice intake, access control, privacy guardrails, PHI de-identification and re-identification, AI draft generation, radiology calculators, RAG retrieval, human approval/adjustment, HTML rendering, Telegram delivery, and credit-based usage.

Roadmap / future scope: dedicated web/mobile application, richer administrative dashboard, direct RIS/PACS integration, additional calculators, stronger idempotent billing controls, formal privacy/security validation, clinical evaluation, and production-grade deployment hardening.

2. Project Planning

2.1 Problem Statement

Radiologists and radiology residents may spend a significant part of their working day manually converting findings into polished reports. The repeated phrasing, formatting, manual measurements/classification checks, and administrative work reduce the time available for image interpretation and clinical reasoning.

RADRAFT addresses this by converting dictated or typed imaging findings into a structured draft while preserving human clinical authority.

2.2 Project Goal

Build a safe, auditable AI radiology reporting assistant that accelerates report drafting without replacing the radiologist's judgment.

2.3 Objectives

- Reduce repetitive report-writing effort.
- Accept both text and voice findings.
- Generate consistent radiology report structure.
- Protect patient identifiers before AI processing.
- Ground supporting knowledge using a RAG pipeline.
- Use deterministic calculators for radiology scores/classifications instead of relying on free-form LLM calculation.
- Require radiologist approval before final delivery.
- Produce a portable HTML report suitable for later transformation into PDF, JSON, XML, or RIS/PACS fields.
- Maintain access control, subscription usage, and audit-friendly workflow boundaries.

2.4 Target Users

Primary users: junior radiologists and radiology residents.

Initial market context in the project presentation: Cairo, Egypt; Ministry of Health, government hospitals, and private hospitals.

2.5 Functional Scope

Area	MVP Requirement	Status
Authentication / access	Validate Telegram user against PostgreSQL	Implemented
Subscription control	Allow processing only when active credits exist	Implemented
Input	Text, voice note, audio	Implemented
Voice transcription	Convert audio to text	Implemented
Guardrails	Harm, prompt injection, topic checks	Implemented
PHI protection	Detect, de-identify, map, re-identify	Implemented
Clinical drafting	Structured AI-generated radiology draft	Implemented
Calculators	Deterministic radiology tools	Partially implemented; expansion planned
RAG	Qdrant retrieval + reranking	Implemented
Human review	Approve or request adjustment	Implemented
Report revision	Apply targeted edits and return new version	Implemented
Rendering	Standalone HTML report	Implemented
Delivery	Telegram document delivery	Implemented
Usage deduction	Deduct one credit after approved delivery	Implemented, with idempotency improvement planned
Web/mobile app	Dedicated UI	Roadmap
RIS/PACS integration	Hospital-system integration	Roadmap

2.6 Work Breakdown Structure (WBS)

Phase A — Discovery & Requirements

- Define radiologist pain points and target reporting tasks.
- Define workflow inputs/outputs.
- Identify clinical safety boundaries.
- Define report structure and acceptance criteria.

Phase B — Data & Access Layer

- Configure PostgreSQL user/subscription data.
- Implement Telegram authorization check.
- Implement session/credit logic.
- Configure doctor metadata retrieval.

Phase C — Privacy & Safety

- Build topic guard and prompt-injection checks.
- Build PHI detection cascade.
- Implement de-identification placeholder mapping.
- Validate safe re-identification boundaries.

Phase D — Clinical Intelligence

- Implement system prompt and structured drafting policy.
- Add deterministic radiology calculators.
- Build RAG ingestion pipeline.
- Configure Qdrant retrieval and Cohere reranking.
- Define citation/grounding behavior.

Phase E — Human Review

- Deliver draft to radiologist.
- Implement approve/adjust control.

- Re-run PHI protection on edit requests.
- Maintain report versions through the revision loop.

Phase F — Rendering & Delivery

- Generate semantic HTML.
- Fill patient/doctor/date fields from trusted workflow data, not model generation.
- Validate required placeholders.
- Deliver final report.
- Deduct report credit after successful delivery.

Phase G — Testing & Hardening

- Workflow-level test cases.
- Guardrail and prompt-injection cases.
- PHI detection cases.
- Calculator correctness cases.
- RAG grounding cases.
- Revision-loop regression cases.
- Credit and authorization edge cases.

2.7 Team Responsibilities

Team member	Primary responsibility	Main workflow
Islam Hegazy	Access & input gateway	Workflow 1
Radwa Zakarya	Privacy & safety gateway	Workflow 2
Reham Metwally	Clinical AI agent & RAG	Workflow 3
Omaima Abdelhaleem	Human approval & revision	Workflow 4
AbdelRahman Diab	Report rendering & delivery; clinical advisor (radiologist)	Workflow 5

2.8 Key Deliverables

- Working n8n workflow chain.
- PostgreSQL-backed access/subscription layer.
- PHI gate and de-identification workflow.
- AI radiology drafting agent.
- Deterministic calculation tools.
- RAG knowledge base in Qdrant.
- Human approval/revision loop.
- Final standalone HTML report.
- Test-case documentation and execution evidence.
- Demo video and GitHub repository.

2.9 Acceptance Criteria

A complete successful report cycle should satisfy all of the following:

6. An unauthorized or inactive user is blocked before expensive AI processing.
7. A valid text or voice request is normalized into the same downstream input format.
8. PHI is removed/replaced before the clinical prompt reaches the model.
9. Relevant scoring/classification tasks use deterministic tools when available.

10. RAG retrieves relevant medical references without including PHI in the retrieval query.
11. The generated report does not invent findings, measurements, or patient identifiers.
12. The radiologist can approve or request corrections.
13. Any correction request is protected by the PHI pipeline again.
14. Final patient and doctor identity fields come from trusted workflow/database data.
15. The final report is delivered only after validation.
16. Usage is deducted only after successful approved-report delivery.

2.10 Major Risks and Mitigation

Risk	Impact	Mitigation
Hallucinated findings or measurements	High clinical risk	Strict prompts, deterministic calculators, RAG grounding, human approval
PHI leakage	Privacy/security incident	Regex + Presidio, de-identification before model ingress, re-identification in trusted stage
Prompt injection	Unsafe model behavior	Input checks, instruction separation, treat retrieved text as data, not instructions
Wrong or stale RAG evidence	Incorrect support	Controlled ingestion, source metadata, registry, reranking, citation policy
Unauthorized access	Privacy/financial risk	PostgreSQL authorization + account status + credit checks
Credit deducted incorrectly	Billing/user trust issue	Deduct after confirmed delivery; planned unique/idempotent report key
Edit introduces new PHI	Privacy risk	Repeat PHI scan/de-identification on every adjustment cycle
Model edits unrelated content	Clinical integrity issue	Targeted-editor prompt; unchanged text preserved; unclear edit marked not applied
Workflow/service failure	Availability issue	Retry controls, fail-fast validation, explicit branch handling, future monitoring

3. Stakeholder Analysis

3.1 Stakeholder Identification

Primary Stakeholders

Radiologists / Radiology Residents

They are the direct users and final clinical decision-makers. Their priorities are speed, accuracy, report consistency, low friction, and ability to override/edit every output.

Patients

Patients are indirect but high-impact stakeholders because their clinical data, privacy, and downstream care may be affected. Their interests include confidentiality, accurate reporting, and human clinical accountability.

Secondary Stakeholders

Hospital / Radiology Department Management

Interested in throughput, standardization, productivity, cost, governance, and adoption.

Hospital IT / Security / Data Protection Teams

Responsible for infrastructure, identity/access, network security, secrets, data residency, auditability, backup, and incident handling.

Clinical Governance / Quality Teams

Interested in validation, report quality, clinical risk, incident review, version control, and human sign-off.

Project Team / Developers

Responsible for workflow implementation, testing, security controls, prompt/tool design, maintenance, and documentation.

DEPI Reviewers / Mentors

Evaluate whether the project satisfies the graduation requirements, demonstrates AI-in-healthcare concepts, and is documented clearly.

AI / API / Infrastructure Providers

External dependencies such as Telegram, model APIs, embedding/reranking providers, and hosting infrastructure. They influence availability, privacy architecture, latency, and cost.

3.2 Power–Interest Matrix

Stakeholder	Power	Interest	Engagement strategy
Radiologists	High	High	Manage closely; continuous testing and feedback
Hospital management	High	High	Demonstrate KPIs, cost, safety and adoption evidence
IT / Security / DPO	High	High	Involve in architecture, privacy, access and deployment reviews
Clinical governance / QA	High	High	Formal validation, incident review and sign-off rules
Patients	Medium	High	Protect privacy; preserve clinician accountability and transparency
Project team	High	High	Clear ownership, version control, testing and issue tracking
DEPI reviewers	High	Medium/High	Provide traceable documentation, demo evidence and requirement mapping
External API providers	Medium	Medium	Monitor service, privacy terms, limits, cost and fallback planning

3.3 Stakeholder Needs

Stakeholder	Main needs
Radiologist	Fast input, minimal clicks, reliable report, transparent classifications, easy correction
Department lead	Standardized output, measurable time saving, controlled adoption
IT/Security	Least privilege, secret isolation, auditability, PHI controls, recoverability
Clinical QA	No autonomous final diagnosis/report release; versioning and review evidence
Patient	Confidentiality and accurate clinician-reviewed output
Finance/Admin	Predictable credit usage and subscription visibility
Development team	Modular workflows, observable failures, repeatable test cases

3.4 Simplified RACI

Activity	Radiologist	Workflow owner	Clinical advisor	IT/Security	Project lead/team
Define report	C	C	A/R	I	R

requirements					
Build workflow	I	R	C	C	A
Validate clinical output	A/R	C	R	I	C
Privacy/security review	I	C	C	A/R	R
Approve final patient report	A/R	I	C	I	I
Deploy/update system	I	R	C	A/R	A/R
Incident review	C	R	C	A/R	A/R

R = Responsible, A = Accountable, C = Consulted, I = Informed.

4. Database Design

4.1 Design Goals

The data layer supports four main concerns:

17. **Identity and authorization** — determine whether the Telegram user is an active authorized radiologist.
18. **Subscription/usage control** — track available report credits.
19. **Report/session traceability** — support report lifecycle, versions and delivery state.
20. **Knowledge-base ingestion state** — avoid duplicate RAG ingestion and maintain document traceability.

The relational database is PostgreSQL. The medical knowledge embeddings are stored separately in Qdrant.

4.2 Verified Current PostgreSQL Data Used by the Workflow

`authorized_users` — Telegram gateway account store

The interactive presentation shows this table being used for the Telegram authorization, trusted doctor signature, and subscription balance. This is the *gateway access model*. It must not be silently equated with the separately discussed `login_id`-based relational account model without checking the deployed schema.

Verified fields referenced by the workflow:

Field	Purpose
`telegram_id`	Primary user identifier used by the Telegram MVP
`full_name`	Trusted doctor name used in the final signature block
`sessions_available`	Remaining report credits
`account_status`	Active/inactive authorization state

Authorization logic: user is accepted only when `telegram_id` matches, `account_status = 'active'`, and `sessions_available > 0`.

Credit update rule: decrement is guarded so the balance cannot become negative, and the new balance is returned after the update.

`RAG_updated_registery`

A registry used by the ingestion workflow to check whether a Google Drive file has already been processed.

Verified field referenced in the presentation:

Field	Purpose
`File_id`	Unique source-file identifier used to prevent duplicate vector ingestion

> Implementation note: the workflow roadmap recommends writing the registry row only after a successful Qdrant upsert so a failed ingestion cannot be incorrectly marked complete.

4.3 Qdrant Vector Collection

`radraft_medical_guidelines`

Used by Workflow 3 for RAG retrieval.

- Clinical reference documents are chunked and embedded.
- Qdrant returns top candidate chunks.
- Cohere reranks candidates before the agent receives the final context.
- Metadata should preserve source title/document identity, section/page where available, and ingestion/version information.
- Retrieval queries should not contain patient PHI.

4.4 PostgreSQL design — findings from our earlier database-building conversation

The database was **implemented and exercised in PostgreSQL**, rather than merely proposed as an ERD. The earlier database conversation describes **six actual tables**, database-side constraints and privilege restrictions. However, the currently available conversation excerpt does **not contain the complete six CREATE TABLE statements or their full column lists**. Accordingly, this submission does not present guessed table names/columns as physical schema facts. The next SQL export is the authoritative way to finish the physical data dictionary.

Verified database test evidence

Evidence from our earlier work	What it establishes	What it does not establish
A test doctor had `id = 1`, `name = Test Doctor`, `email = test.doctor@example.com`, `login_id = rd_test_doctor_001`	Separate login-based test account exists; a dummy email was added for end-to-end testing	Table name and full column constraints not visible in the excerpt
Test SQL output: `new_report_id = 6`, `new_session_id = 6`, `quota_used_now = 4`, `quota_limit_now = 100`	A report-session creation path returned IDs and quota usage values during testing	Does not prove billing-on-final-delivery or that IDs must always match
`start_report_session(bigint,text,bigint)` owned by `radraft_owner`, `SECURITY DEFINER = true`, `n8n_can_execute = false`	Old three-argument signature exists but n8n execution was denied in the inspected state	Does not establish that overload can safely be removed
`start_report_session(bigint,text,text,bigint)` owned by `radraft_owner`, `SECURITY DEFINER = true`, `n8n_can_execute = true`	Four-argument overload is executable by the n8n database role in the inspected state	Does not prove permissions on every other function or table

Important distinction: The presentation's `telegram_id`/`sessions_available` subscription gateway and the database discussion's `login_id`/report-session/quota function should be documented as **two observed interfaces/data models** until the team demonstrates the actual join or mapping between them. In particular, `quota_used_now` / `quota_limit_now` and `sessions_available` are not assumed to be the same counter.

Function interface and least-privilege boundary

```
-- Inspected in our database chat (signatures only; not implementation bodies)

start_report_session(bigint, text, bigint)

start_report_session(bigint, text, text, bigint)
```

The implementation used a function-level **SECURITY DEFINER** boundary and exposed the four-parameter call to the `n8n` role while not exposing the three-parameter overload. This is relevant to the security architecture: workflow credentials should be given only the minimum permitted stored-procedure entry points. Function execution privileges should be verified after each migration; SECURITY DEFINER alone is not a substitute for least privilege.

Logical entity relationships — illustrative, not a claim about physical names

```
erDiagram
    DOCTOR_ACCOUNT ||--o{ REPORT_SESSION : initiates
    REPORT_SESSION ||--o{ REPORT_RECORD : tracks
    REPORT_RECORD ||--o{ REPORT_REVISION : may_have
    DOCTOR_ACCOUNT ||--o{ QUOTA_TRANSACTION : consumes
    REPORT_RECORD o|--o{ DELIVERY_EVENT : produces
```

These are **conceptual** entities capturing the project's observed behavior. The physical ERD must be generated from the actual six-table schema; do not copy these generic names into migration scripts.

SQL checklist to export the real six-table database design

Run the following inside an authorized development instance (no production credentials or patient records in the submission):

```
-- Physical table list
SELECT table_schema, table_name
FROM information_schema.tables
WHERE table_schema = 'public' AND table_type = 'BASE TABLE'
ORDER BY table_name;


-- Authoritative columns, types, nullability and defaults
SELECT table_name, ordinal_position, column_name, data_type,
       is_nullable, column_default
FROM information_schema.columns
WHERE table_schema = 'public'
ORDER BY table_name, ordinal_position;


-- PK / FK / UNIQUE / CHECK constraints
SELECT conrelid::regclass AS table_name,
       conname AS constraint_name,
       contype AS type, pg_get_constraintdef(oid) AS definition
FROM pg_constraint
WHERE connamespace = 'public'::regnamespace
ORDER BY conrelid::regclass::text, conname;


-- Index definitions
SELECT tablename, indexname, indexdef
FROM pg_indexes WHERE schemaname = 'public'
ORDER BY tablename, indexname;


-- Function owners, SECURITY DEFINER and n8n execute grants
SELECT p.oid::regprocedure AS signature,
       pg_get_userbyid(p.proowner) AS owner,
       p.prosecdef AS security_definer,
       has_function_privilege('n8n', p.oid, 'EXECUTE') AS n8n_can_execute
```

```
FROM pg_proc p
JOIN pg_namespace ns ON ns.oid = p.pronamespace
WHERE ns.nspname = 'public'
ORDER BY p.oid::regprocedure::text;
```

Once the team exports these results, replace the conceptual ERD with the actual six-table ERD, along with PK/FK cardinalities, checks, indexes, and the precise quota transaction rules.

4.5 Database Design Principles

- **Parameterised SQL:** no direct string concatenation for user identifiers.
- **Least data exposure:** only required doctor/account metadata should leave the DB node.
- **Atomic credit update:** decrement with a conditional guard and `RETURNING` result.
- **Idempotency:** production billing should use a unique report/delivery key so retries cannot charge twice.
- **Auditability:** important state transitions should be timestamped and attributable.
- **Separation of concerns:** relational identity/report state in PostgreSQL; semantic medical knowledge in Qdrant.
- **Trusted identity:** final doctor name must come from PostgreSQL, not the LLM.
- **No PHI in vector queries:** patient identifiers must not be embedded into RAG search text.

4.6 Database Security Controls

- Database credentials remain inside n8n credentials/secrets management.
- Application queries use minimum required privileges.
- Patient identifiers should not be stored unnecessarily in logs.
- Database/network access should be restricted to the orchestration environment.
- Backups and retention rules should be defined before real clinical deployment.
- n8n's own internal database is currently separate in the MVP; moving it to PostgreSQL is listed as a production-hardening step.

5. UI/UX Design

5.1 UX Objective

The interface should let a radiologist create a report with the fewest possible steps while keeping safety-critical states explicit. The MVP uses Telegram because it provides a familiar, low-friction interaction model and supports text, voice notes, documents, buttons/forms, and delivery notifications.

5.2 Primary Persona

Junior Radiologist / Resident

Context: high case volume, frequent repetitive report wording, limited time.

Goal: convert findings into a structured draft quickly.

Needs: text or voice input, confidence that measurements/classifications are handled correctly, easy correction, clear approval step, and final portable report.

Pain points: typing burden, repeated formatting, manual calculations, switching between references, and late documentation work.

5.3 Core User Journey

```
flowchart LR
    A[Open RADRAFT Telegram bot] --> B[Send findings by text or voice]
    B --> C{Authorized & credits available?}
    C -- No --> D[Show access/subscription message]
    C -- Yes --> E[Safety + PHI processing]
    E --> F[Generate structured draft]
    F --> G[Receive draft file]
    G --> H{Approve or Adjust?}
    H -- Adjust --> I[Send requested correction]
    I --> J[Receive revised version]
    J --> H
    H -- Approve --> K[Render final HTML]
    K --> L[Deliver final report]
    L --> M[Show remaining credits]
```

5.4 MVP Screens / Interaction States

State 1 — Input

- User sends typed findings, voice note, or audio.
- No complex form is required at the MVP stage.
- The interaction should preserve natural radiologist dictation style.

State 2 — Authorization Feedback

- Authorized: short status such as “Authorization granted. Checking input...”
- Unauthorized/inactive/no credits: stop processing and display a clear support/subscription message.

State 3 — Processing

Recommended UX behavior:

- Use concise progress messages only at meaningful transitions.
- Avoid exposing internal model/tool logs.
- Do not display PHI-detection internals to the user unless intervention is required.

State 4 — Draft Review

- Send the complete draft as a document so formatting is readable.
- Present two obvious choices:
- ✅ **Approve**
- ✏️ **Adjustment**
- The user must always retain final control.

State 5 — Adjustment

- User describes only the requested correction.
- System applies targeted edits and sends a new version.
- The loop continues until approval.

State 6 — Final Delivery

- Generate a standalone HTML report.
- Include trusted patient/doctor/date fields.
- Send with a clear caption such as “Final Approved Radiology Report”.
- Only after confirmed delivery should the usage credit be deducted.

5.5 UI/UX Principles

1. Minimal cognitive load

The radiologist should mainly perform three actions: **send findings → review → approve/adjust**.

2. Human primacy

The interface must never make the AI draft look like a silently finalized report. Approval is an explicit action.

3. Progressive disclosure

The user sees the information needed for the current step. Technical details such as embeddings, reranking, PHI mapping and model routing stay behind the interface.

4. Error visibility

Failures should produce clear, actionable messages: unauthorized account, no credits, unsupported input, transcription failure, processing failure, or delivery failure.

5. Consistent terminology

Use the same terms throughout: **Draft Report**, **Adjustment**, **Approved Report**, **Reports Remaining**.

6. Safety by design

- No final report without human approval.
- Patient identity fields are inserted from trusted workflow data.
- PHI protection repeats during the edit loop.
- The UI never encourages the user to treat the draft as autonomous medical advice.

5.6 Final Report UX

The HTML report should be optimized for both screen viewing and printing:

- Semantic sections for examination/findings/impression/recommendations as applicable.
- Clear patient information header.
- Consistent typography and spacing.
- A4 print rules.
- Signature block kept together.
- Mobile-responsive display.
- No JavaScript, external fonts, or external images.
- Required identity placeholders inserted exactly once by trusted code.

5.7 Accessibility & Localization Considerations

For a future web/mobile version:

- Support Arabic and English interface text.

- Preserve standard English radiology terminology inside professional reports unless localization is explicitly requested.
- Use high-contrast status indicators that do not rely on color alone.
- Use large tap targets for Approve/Adjustment actions.
- Ensure keyboard accessibility in a web UI.
- Add clear loading, success, warning and error states.

5.8 Future UI Roadmap

21. **Web dashboard** for secure login, report history, search and review.
22. **Admin dashboard** for user provisioning, subscriptions and usage monitoring.
23. **Structured examination templates** for common modalities and body regions.
24. **RIS/PACS integration** so selected patient/study context can be bound to the reporting session.
25. **Report history and version comparison** with an audit trail.
26. **Analytics dashboard** showing usage, turnaround time, adjustment rate and failed runs.
27. **Institution-specific templates** and branding.

6. Submission Structure Recommendation

For the required **Documentation link**, publish this documentation as one GitHub Pages / GitHub Markdown / Notion / Google Docs landing page with the following navigation:

28. **Project Overview**
29. **Project Planning**
30. **Stakeholder Analysis**
31. **Database Design**
32. **UI/UX Design**
33. **System Architecture**
34. **Testing & Safety**
35. **GitHub Repository**
36. **Demo Video**

Recommended Evidence to Embed

- System architecture diagram from the presentation.
- Screenshots of the five main n8n workflows.
- PostgreSQL ERD or schema export.
- Telegram interaction screenshots: input, review, adjustment, approval, final report.
- One sample final HTML report using synthetic patient data.
- Test-case sheets for workflows and guardrails.
- GitHub repository link and demo video link.

7. Final Submission Note

RADRAFT should be presented as an **AI-assisted radiology reporting system with explicit human approval**, not as an autonomous diagnostic product. The current implementation is an educational MVP and proof of concept. Production or real-patient deployment would require formal clinical validation, privacy/security assessment, institutional governance, regulatory review where applicable, stronger monitoring, and production-grade infrastructure.

8. Cross-checked evidence from our actual project conversations

This chapter is the difference between this revision and a generic proposal document. It tracks what the team *actually discussed, created, tested or debugged*, and avoids advertising unverified claims as production guarantees.

8.1 Conversation-based development history

Work thread	Discussion and artifacts	Documentation consequence
Radiology Assistant MVP and business model	AI-assisted radiology report drafting, radiologist target users, time-saving value proposition, longer-term commercial viability	Scope and stakeholder rationale
Brand and storytelling	RADRAFT product name, logo/slogan, elevator pitch, executive infographic, technical blueprint	Product identity, positioning and roadmap
n8n gateway and database construction	Telegram workflow, PostgreSQL authorization, six-table DB and `start_report_session` function, test doctor identity and quota outputs	Physical DB discussion and workflow interfaces
Clinical prompt/guardrail tuning	Topic guard that accepts medical shorthand, Arabic/English mixed findings, patient header placeholders and classification requests; debugging false rejections	UX error taxonomy and safety test cases
Clinical edge-case reports	Breast ultrasound with suspicious lesion descriptors; prostate mpMRI with PI-RADS-relevant features were incorrectly rejected by the on-topic classification	Verified issue to test, **not** proof of clinical accuracy
Workflow export and integration	Report creation/delivery, database ingestion and clinical AI/RAG n8n JSON exports discussed, plus per-workflow test-case spreadsheets	QA traceability and evidence structure
HTML report formatter	Explicit preserve-all-findings prompt, no invented measurements/classifications, no generic normal sections, no fabricated reporting credentials	Reporting design constraints and output QA
Graduation submission review	Team already had a GitHub repository link and demo video, then asked to compare against DEPI graduation guidelines	Submission checklist should link actual artifacts rather than invent URLs

8.2 Workflow-by-workflow technical traceability

ID	Trigger / flow	Decisions and data boundaries	Evidence needed for final submission
WF1	Telegram trigger → check `is_bot` → authorized doctor and credits → detect text/voice/audio → transcribe voice → fetch doctor identity → WF2	Reject unrecognized/unauthorized senders and unsupported media before any expensive LLM processing	Redacted Telegram flow screenshot; JSON export; access test cases
WF2	Harmful content check → injection/NSFW check → radiology topic guard → regex identifiers → Presidio → de-identifiers → WF3	Maintain placeholders; do not send names/identifiers to agent; preserve clinically meaningful terminology	PHI redaction tests, Arabic/English on-topic tests, attack tests
WF3	Clinical agent → tool calls → Qdrant retrieval and Cohere reranking → report draft → controlled restoration → WF4	Calculator result must be tool-derived; source retrieval must be traceable; no cross-user context leakage	Tool traces, RAG retrieval examples, structured-output tests
WF4	Draft TXT to radiologist → approve or adjust → targeted	Approval required; preserve unrelated findings; repeat	Telegram screenshots showing review, revision and approval

	edit loop → approved report	safety controls for edit requests	
WF5	Trusted metadata + report content → HTML render and validation → Telegram delivery → credit workflow	Distinguish successful send from generation; only approved version becomes final	Sanitized HTML screenshot, delivery trace, billing test
Support: Knowledge ingestion	Google Drive PDFs → registry lookup → extraction → recursive splitting → embedding → Qdrant	Avoid duplicate indexing; record source identity and versions	Ingestion workflow export + registry/output screenshots
Support: Account management	Manual administrative upsert of doctor account/credit balance	Admin-only mutation, least privilege, use synthetic identities for demos	Redacted SQL + test account evidence

8.3 Design parameters observed in the interactive technical presentation

Component	Observed configuration	Interpretation / caution
WF2 local topic guard	Ollama `qwen2.5:0.5b`, PASS-when-uncertain behavior, low threshold	Configuration described, not an established clinical sensitivity/specificity
PHI stage	Regex plus Microsoft Presidio analyzer	Does not guarantee identification of every indirect identifier
WF3 connected calculators	`ti_rads`, `bi_rads`, `psa`, `pi_rads_tz`, `pi_rads_pz`	Five connected tools in the walkthrough; additional scoring systems are future scope
RAG	OpenAI `text-embedding-3-large` (3072-dim), Qdrant 10 candidates, Cohere reranking to 5	Dimensions must match collection; retrieval does not itself guarantee clinical correctness
Document ingestion	Recursive text splitter chunk size 3000, overlap 400; embeddings batched	PDF extraction may fail on diagrams, tables and scans
Report output	Standalone semantic HTML, patient/doctor metadata sourced from trusted workflow data	HTML delivered for review/integration; not a production RIS/PACS connector
Subscription	Telegram user `sessions_available`, guarded decrement after approved delivery	Retries/double charging still require idempotency proof

8.4 Confirmed issues and regression test cases

Test ID	Scenario from project work	Required result	Status shown by conversation
REG-01	Breast US description: irregular hypoechoic non-circumscribed lesion, vascularity, measurements	Topic guard must recognize a valid radiology report request; proceed to downstream validation	**Previously rejected — regression required**
REG-02	Prostate mpMRI with PSA 9.2, 14-mm right PZ lesion, restricted diffusion and DCE	Guard must allow medically appropriate MRI descriptors and PI-RADS-related reasoning	**Previously rejected — regression required**
REG-03	Mixed Arabic/English medical wording or PHI placeholders	Pass on-topic clinical text, reject only clearly unrelated messages or instruction attacks	Prompt policy designed; automated pass results not established
REG-04	Word “lobulated” used for thyroid nodule margins	Descriptor must not be redacted as a `PERSON` and lost before TI-RADS tool evaluation	Earlier bug/fix described in technical walkthrough; independent retest required
REG-05	De-identified synthetic test patient name, DOB and MRN	Identifier values replaced in model input and safely restored only in trusted output	Pipeline design described; formal PHI benchmark not provided
REG-06	Revision request alters one fact only	Apply targeted edit, keep all unrelated report information unchanged	Explicit editing requirement; execution evidence needed
REG-07	Rendering a report with missing demographic information	Do not fabricate headers or default normal findings; fail visibly if required metadata is missing	Formatter contract defined; full test evidence needed
REG-08	Duplicate delivery/retry	Do not double-charge credits for a single approved report	**Hardening gap / must be tested**
DB-01	Start a report session with test	Returns created IDs and quota	Observed output (IDs 6/6, used

	doctor	consumption within configured maximum	4, limit 100)
DB-02	n8n execution of stored function	Allowed only for specifically granted signature	Observed on four-argument overload only

8.5 Team ownership (from the technical walkthrough)

Owner	Area
Islam Hegazy	Workflow 1: Telegram access and normalization; topic-guard test discussions
Radwa Zakarya	Workflow 2: privacy, PHI gate and guardrails; relational database planning discussions
Reham Metwally	Workflow 3: clinical AI agent and RAG
Omaima Abdelhaleem	Workflow 4: human review and revision
AbdelRahman Diab	Workflow 5: report rendering and delivery; radiology subject-matter input

These names and associations are based on the provided technical presentation and project discussion, not a formal employment or assignment record.

9. UI/UX deliverables: current Telegram interface versus design documentation

9.1 MVP interaction map (as built/described)

```
stateDiagram-v2
  [*] --> MessageReceived
  MessageReceived --> AccessDenied: bot / inactive / zero credits
  MessageReceived --> InputType: authorized
  InputType --> Unsupported: other media
  InputType --> Normalize: text
  InputType --> Transcribe: voice or audio
  Transcribe --> Normalize
  Normalize --> SafetyGate
  SafetyGate --> Rejected: off-topic / injection / unsafe
  SafetyGate --> Drafted: accepted and drafted
  Drafted --> Adjustment: doctor asks for change
  Adjustment --> Drafted: updated revision
  Drafted --> Approved: doctor approves
  Approved --> Delivered: render and send HTML
  Delivered --> CreditUpdate
  CreditUpdate --> [*]
```

9.2 Required screen/evidence inventory

For submission, capture **real screenshots** of these Telegram states (with synthetic or redacted clinical data): initial welcome/authorized message; unsupported media; unauthorized/zero-credit message; text and voice inputs; processing feedback; draft TXT; approve/adjust controls; edited draft; final HTML document; remaining-credits notification.

The UX must distinguish a **draft** from a **final approved report** visibly, not merely in internal workflow variables. The physician should be able to recover from transcription issues or guardrail false positives without entering a broken session.

9.3 Explicit UX decisions from our conversations

- **Doctor-first channel:** Telegram is the current interaction, not a bespoke web application.
- **Mixed-language intake:** English, Arabic and common radiology shorthand may be present in the same request. The final report in the current technical prompt is professional English.
- **Report integrity:** Preserve every supplied observation, laterality, measurement, classification and relevant negative finding; never invent a normal finding or a credential.
- **Review action:** Doctor approves or requests a specific adjustment before final release.
- **Readable output:** Semantic headings, findings, impression and optional source-derived patient/doctor header, styled for printing and potential downstream conversion.
- **Transparent refusal:** An on-topic input should not be rejected merely because it contains suspicious-looking clinical words; block truly unrelated or malicious requests and record test evidence for false positives.

9.4 Future UI/UX design — not part of the completed MVP

A dedicated responsive web portal, authenticated dashboard, report history, template library, usage balance and RIS/PACS export have appeared in the product roadmap. They are **design directions**, not implemented screens. Do not mix future-state wireframes with screenshots of the shipped Telegram bot.

10. Deliverable checklist and publication instructions

Required documentation item	What this file covers	Recommended supporting artifact
Project Planning	problem, objective, MVP scope, WBS, responsibilities, acceptance criteria, risks and limitations	Gantt chart with **actual** dates/owners from team history; GitHub milestones
Stakeholder Analysis	primary/secondary stakeholders, power-interest matrix, RACI, needs	stakeholder map figure; validated interviews if available
Database Design	gateway store, Qdrant collection, confirmed function permissions/session test; conceptual relationships	**essential:** physical ERD exported from six real tables + schema SQL + constraints
UI/UX Design	persona, Telegram journey, state diagram, review/revision, errors, report layout	anonymized **real** Telegram screenshots and final HTML specimen
Testing evidence	actual debug cases, regression requirements and test inventory	four/five workflow XLSX files, guardrail workbook, redacted n8n execution runs
Technical implementation	node-level workflow trace, models, data flow, security boundaries	n8n JSON exports, clean architecture diagram
Project demo	test walkthrough and human signoff	add the team's **actual** GitHub and demo-video links

Publishing a single Documentation link: Commit this Markdown file into the project repository as `docs/RADRAFT_Project_Documentation.md`, add its link in the repository `README.md`, and ensure that the reviewers can access the repo. Alternatively publish the same content on a share-enabled Notion/Google Docs page. The present local file is not itself a publicly accessible Documentation URL.

Do not put credentials, real patient identifiers, private physician contact details or unsanitized execution payloads into the public repository.

10.1 Outstanding confirmations before final submission

37. Export the actual six PostgreSQL table names, fields, keys, checks, indexes and relationships; reconcile the Telegram-credit model with the login/session quota model.
38. Verify the current executable version of each n8n workflow, because a slide describing a node is not a live execution trace.
39. Add actual UI screenshots and final report sample with synthetic data.
40. Attach the already-prepared per-workflow Excel test matrices and mark test outcomes truthfully.
41. Insert actual GitHub/demo links and real milestone dates; they are not supplied in the currently available excerpts.
42. State privacy, patient-safety, accuracy and cost results only if measured; do not call the prototype clinically validated.